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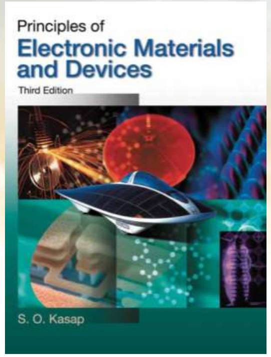

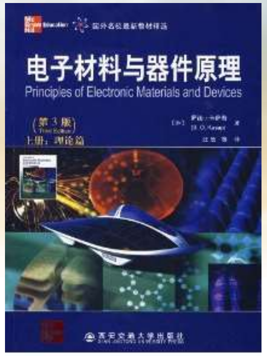

《固态电子学》
Solid-State Electronics

Ch3-1
Elementary Quantum Physics

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Some slides are modified from other Profs’ PPTs. Their contributions are greatly acknowledged.

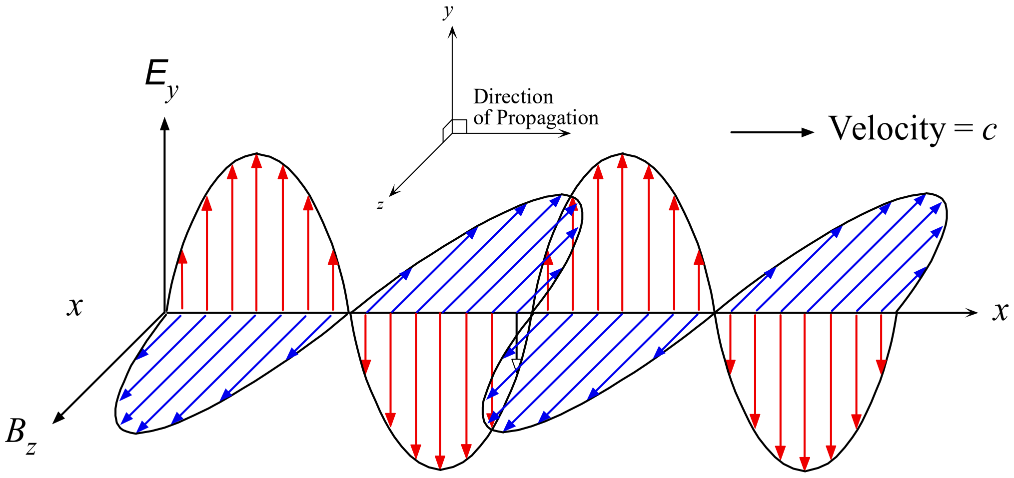
Why need Quantum Physics?

- Classical physics (Newton physics, Maxwell physics) cannot explain lots of experimental phenomena especially electron behaviors when research go into submicron area (atomic level).
  - ✓ Light is a wave or a particle?
  - ✓ Electron is a particle or a wave?

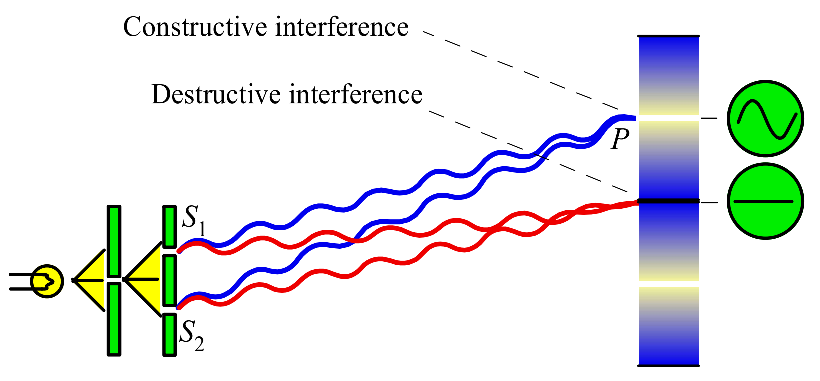
- In submicron area (atomic level), we need quantum physics to explain experimental phenomena and predict electron behaviors.

- Quantum mechanics is
  - ✓ the set of rules obeyed by small systems molecules atoms, and subatomic particles.
  - ✓ One of the two greatest achievements of 20^{th} century physics.
  - ✓ the fundamental principle of semiconductor and photonics. The basis for new research into smaller electronic devices (e.g., quantum dots)

Evidence for Wave-Nature of Light



The classical view of light as an electromagnetic wave. An electromagnetic wave is a travelling wave which has time varying electric and magnetic fields which are perpendicular to each other and to the direction of propagation.



Schematic illustration of Young's double slit experiment.

$$E_y(x,t) = E_0 \sin(kx - \omega t)$$

$$S_1P - S_2P = n\lambda$$

$$S_1P - S_2P = \left(n + \frac{1}{2}\right)\lambda$$

Classical electrodynamics unalterably leads to the picture of a continuous transfer of energy <u>by way of electromagnetic waves.</u>

Light exhibits diffraction and interference phenomena that are only explicable in <u>terms of wave properties.</u>

Evidence for Wave-Nature of Light

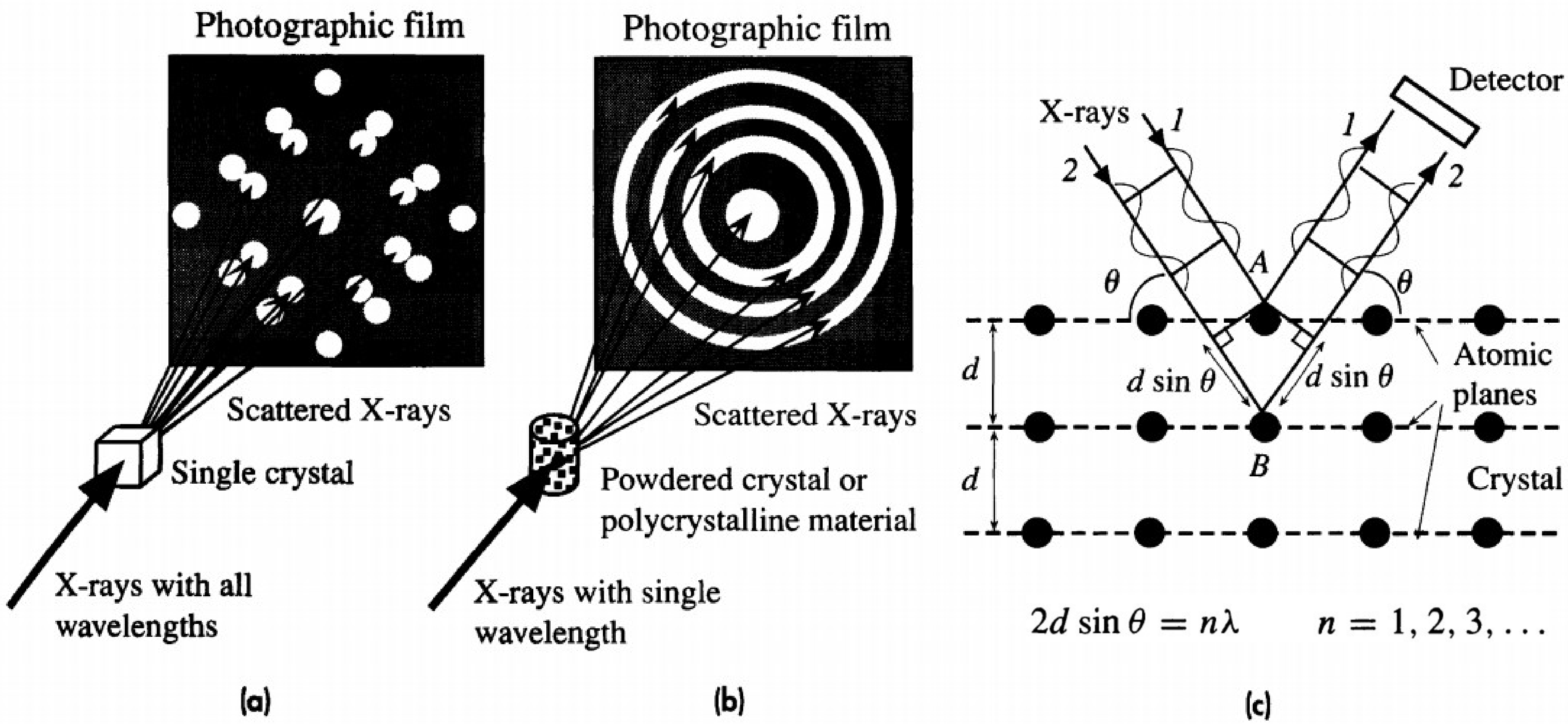


Figure 3.3 Diffraction patterns obtained by passing X-rays through crystals can only be explained by using ideas based on the interference of waves.

(a) Diffraction of X-rays from a single crystal gives a diffraction pattern of bright spots on a photographic film.

(b) Diffraction of X-rays from a powdered crystalline material or a polycrystalline material gives a diffraction pattern of bright rings on a photographic film.

(c) X-ray diffraction involves the constructive interference of waves being "reflected" by various atomic planes in the crystal.

Evidence for Particle-Behavior of Light

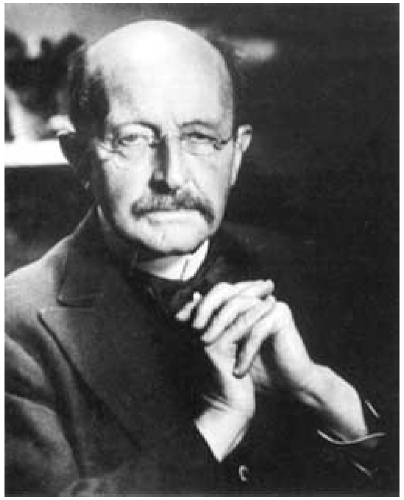
In contrast, the more modern view of Quantum Electrodynamics describes electromagnetic interactions and the transport of energy in terms of massless elementary “particles” known as **photons**. Electromagnetic radiant energy is created and destroyed in quanta or photons, and not continuously as a classical wave.

The History

Some theorists were on the right track, especially Plank, who proposed that nature acted by using "quanta" of energy. But it was the young, unknown Albert Einstein who explained everything and started the field of quantum mechanics with his paper on the photoelectric effect (1905).

Einstein showed that light does not consist of continuous waves, nor of small, hard particles. Instead, it exists as bundles of wave energy called photons.

Blackbody Radiation
Photoelectric Effect
Campton Scattering

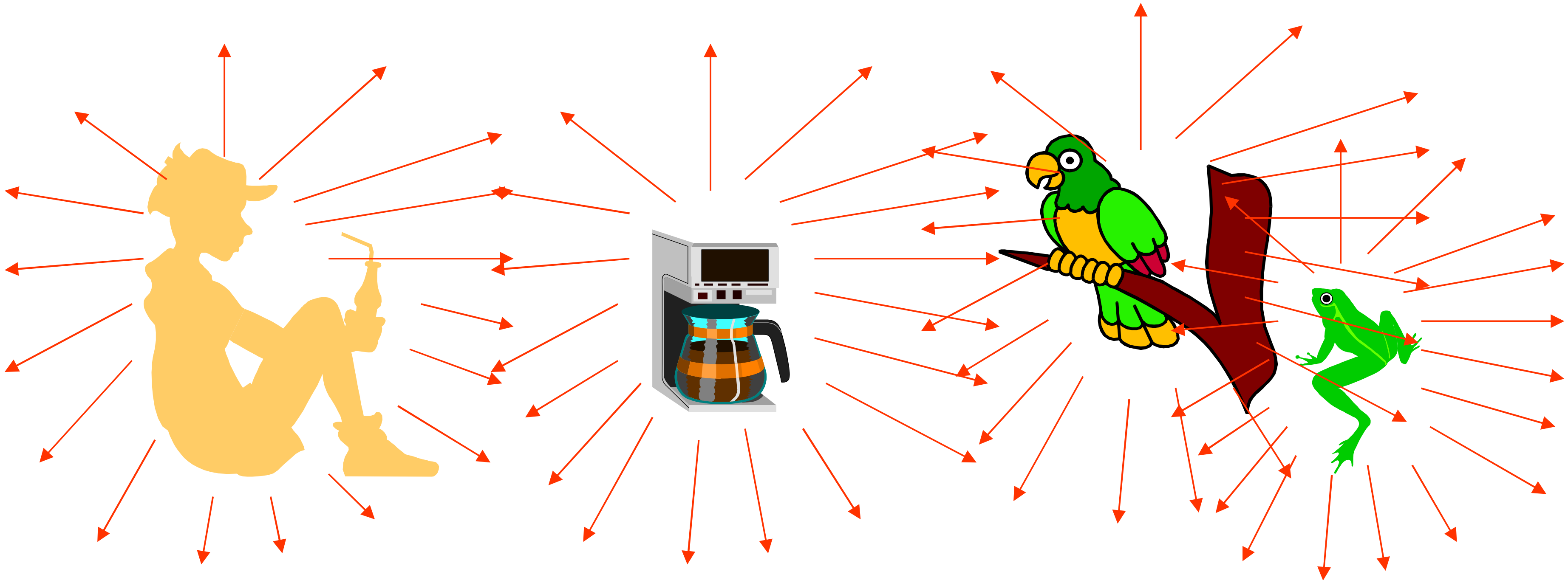


Plank

Albert Einstein

Blackbody Radiation (1900)

Heat Radiation



In 1859, theoretical research of thermal radiation by Gustav Kirchhoff

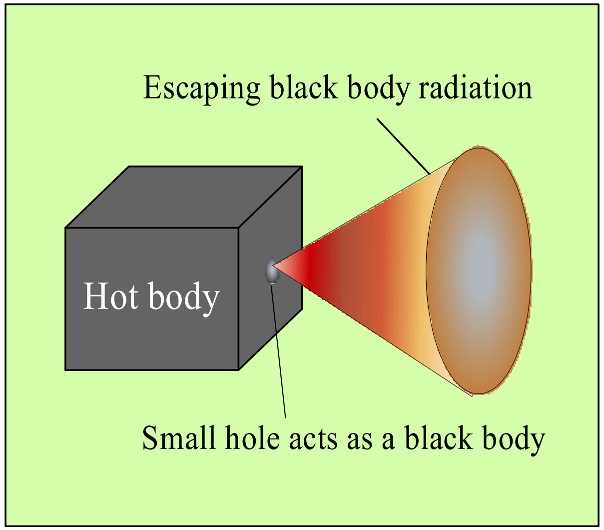
→ Concluded all objects emit and absorb energy in the form of radiation, and the intensity of this radiation **depends on** the radiation wavelength and temperature of the object.

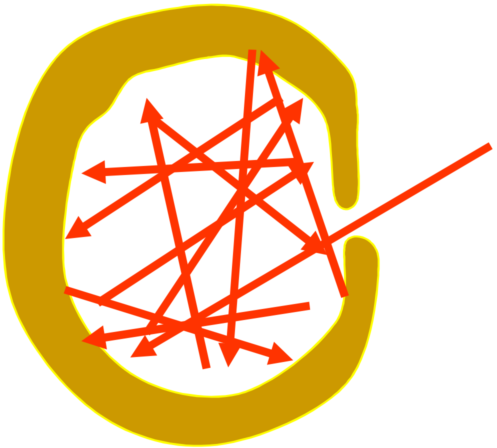
Blackbody Radiation (1900)

Blackbody

Absorption coefficient:
Mirror 0.02; Metal 0.6-0.8; Black particle: 0.95

Let us construct cavity in a metal block, through the wall of which a small hole is drilled. This small hole is called a black body. (a=1)



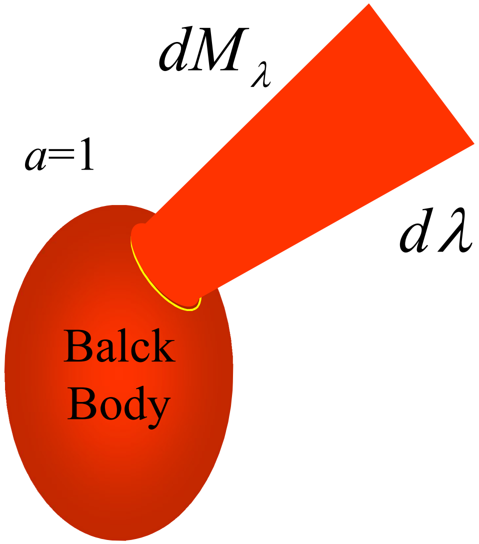


The maximum amount of radiation energy that can be emitted by an object is called the Blackbody Radiation.

Blackbody Radiation (1900)

a) Unit Spectral Irradiance (单色辐出度)

The quantity $M_{(\lambda T)}$ is the rate at which energy is radiated per unit time, per unit wavelength and per unit area of surface for wavelengths lying in the interval λ to $\lambda+d\lambda$ at the temperature T.



$$M_{(\lambda.T)} = \frac{dM_{\lambda}}{d\lambda} \quad (\mathrm{Wm}^{-3})$$

It is a function of wavelength and temperature.

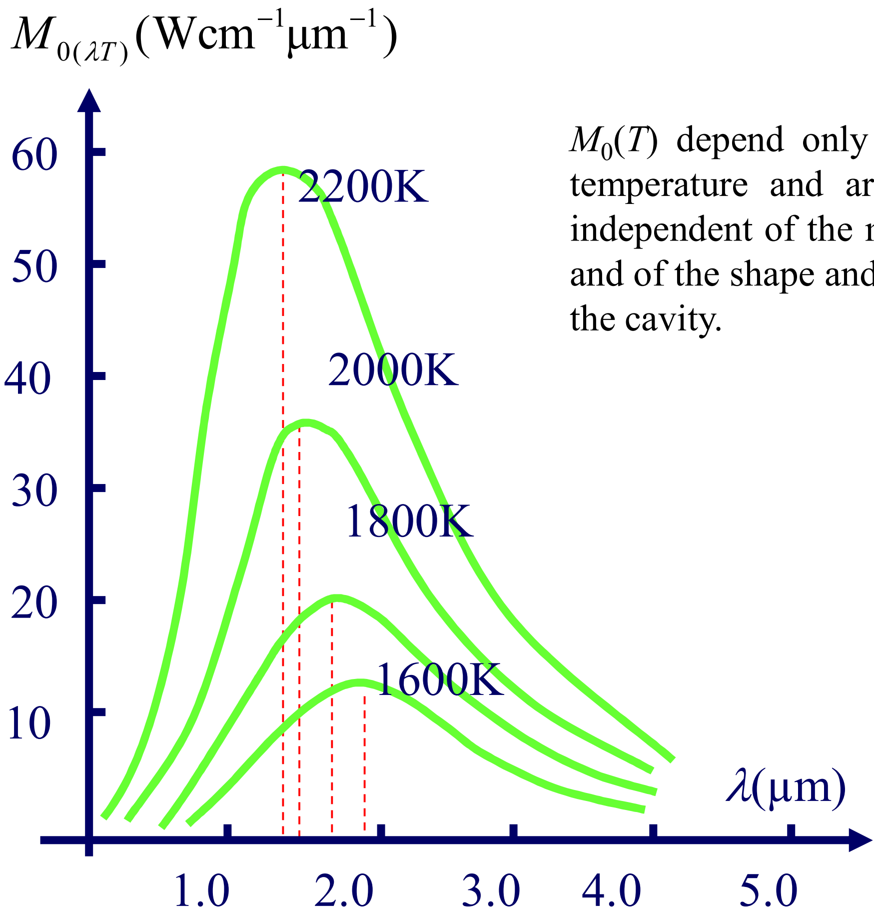
b) Spectral Irradiance (辐出度)

The appropriate quantity $M_{(T)}$ is the rate at which energy is radiated per unit time and per unit area of surface at the temperature T.

$$M_{(T)} = \int_0^{\infty} M_{(\lambda.T)} d\lambda$$

Blackbody Radiation (1900)

Characteristics of Blackbody Radiation



$M_0(T)$ depend only on the temperature and are quite independent of the material and of the shape and size of the cavity.

a) J.Stefan-Ludwig Boltzmarn Law

$$M_{0(T)} = \sigma T^4$$

$$\sigma = 5.67 \times 10^{-8}\ \mathrm{W/mK}$$

J.Stefan—Ludwig Boltzmarn constant

b) Wien Displacement Law

$$T\lambda_m = b$$

$$b = 2.898 \times 10^{-3}\ \mathrm{mK}$$

λ_m： wavelength for peak value

T： absolute temperature

https://phet.colorado.edu/sims/html/blackbody-spectrum/latest/blackbody-spectrum_en.html

Blackbody Radiation (1900)

A number of capable physicists advanced theories based on classical physics, which, however, had only limited success.

Example 2:

Rayleigh-jean's formula

$$M_{0(\lambda T)} = \frac{2\pi CKT}{\lambda^4}$$

C speed of light

K=1.380658×10^{-23}J/K

Boltzmann's constant

Rayleigh-jean's Formula isn't fit the experimental points at short wavelength.

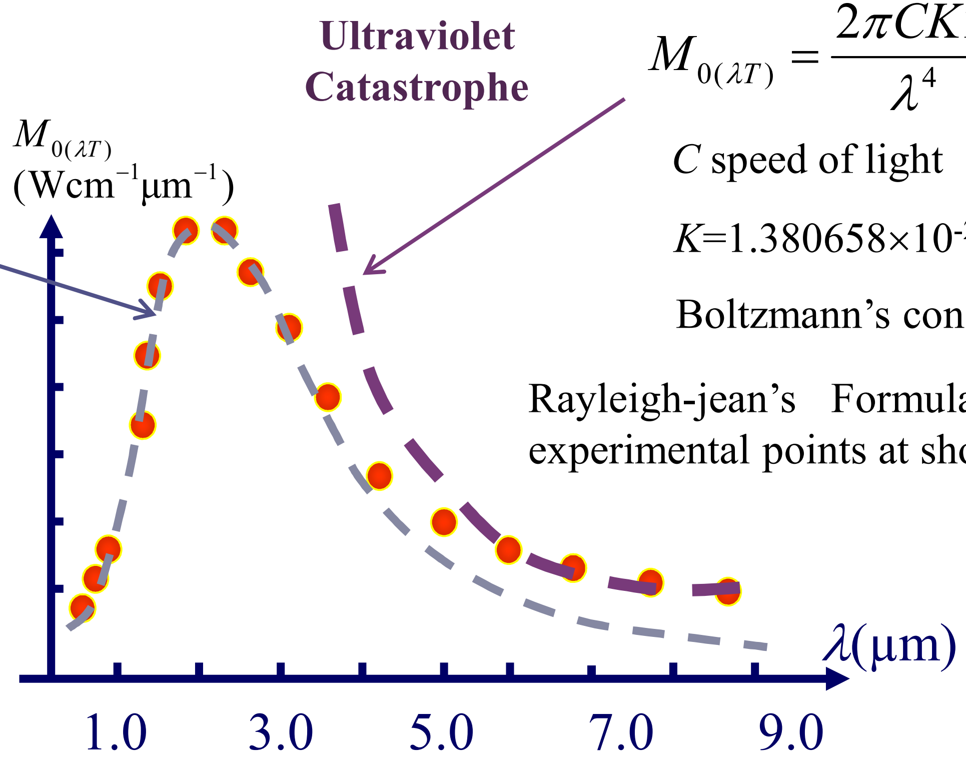
Example 1:

1896, Wien's formula

$$M_{0(\lambda T)} = \frac{c_1}{\lambda^5} e^{-\frac{c_2}{\lambda T}}$$

Where C_1 and C_2 are constants that must be determined by the experimental data.

Wien's formula isn't fit to the experimental points at long wavelength.



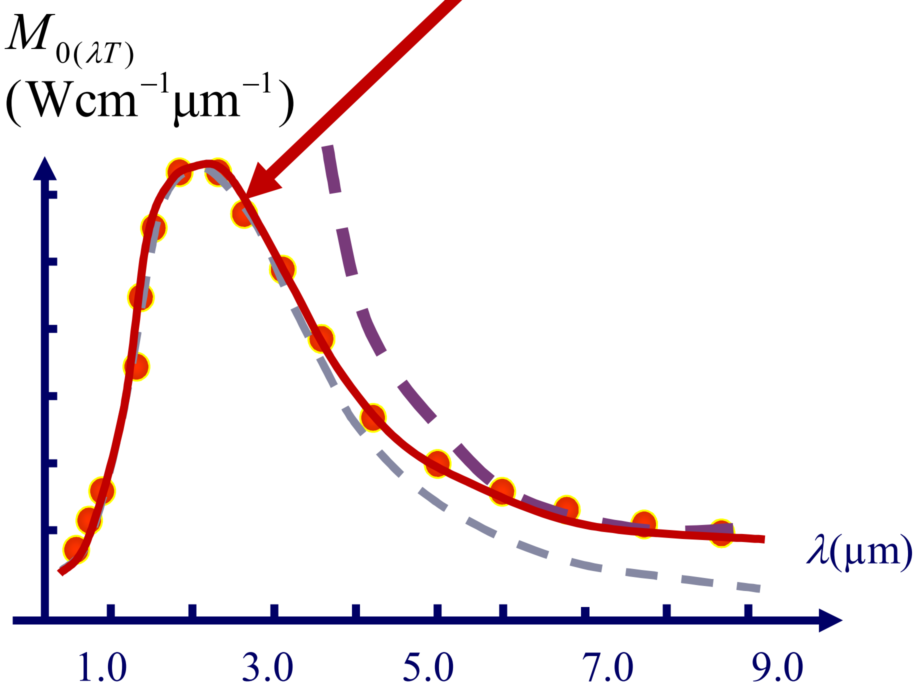
Blackbody Radiation (1900)

In 1900 Max Planck pointed out that if Wien's formula were modified in a simple way it would prove to fit the data precisely. It is

$$M_{0(\lambda T)} = 2\pi h C^2 \lambda^{-5} \frac{1}{e^{\frac{hC}{k\lambda T}} - 1} = 2\pi h C^2 \lambda^{-5} \frac{1}{e^{\frac{h\nu}{kT}} - 1}$$

k: Boltzmann's constant

h=6.63×10^{-34}(js) is called Planck's constant.



Planck's formula fit the data precisely.

Planck's Quantum Supposition

The atoms that make up blackbody walls behave like tiny electromagnetic oscillators. An oscillators can't have any energy but only energies given by

$$E_n = n\varepsilon \qquad n = 1.2.3\cdots$$

where n is a quantum number,

$$\varepsilon = h\nu \qquad h = 6.63\times 10^{-34}\ \mathrm{js}$$

Blackbody Radiation (1900)

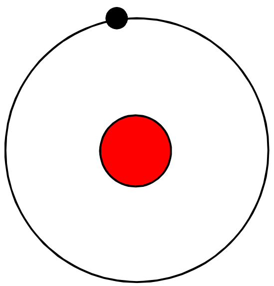
$$I_\lambda = \frac{2\pi hc^2}{\lambda^5\left[\exp\left(\frac{hc}{\lambda kT}\right)-1\right]}$$

We take the equation. One setp further and derive Stefan's Blackbody Radiation Law, to obtain the total radiative power Ps emitted by a blackbody per unit surface area at a temperature T.

$$P_s = \int_0^\infty I_\lambda d_\lambda = \left(\frac{2\pi^5 k^4}{15c^2h^3}\right)T^4 = \sigma_s T^4$$

$$\sigma_s = \frac{2\pi^5 k^4}{15c^2h^3} = 5.670\times10^{-8}\,\mathrm{W\,m^{-2}K^{-4}}$$

Exercises: calculate the energy emitted by the sun and the energy received by the earth per unit area, given that the temperature of the sun is 5800 K, the radius of the sun is $6.96*10^8$ m, the distance between the sun and the earth is $1.496*10^{11}$ m



$$F_0 = \sigma T_0^4$$

$$F(d) = F_0\frac{4\pi r_0^2}{4\pi d^2} = F_0\left(\frac{r_0}{d}\right)^2$$

Answer: 1367 W/m^2

Generally, people define AM1.5=1000 W/m^2 as standard condition to measure the efficiency of solar cell. For example, Solar cell efficiency ~20%, hence 200W/m^2

AM: air-mass（大气质量）

AM1.5 is the distance of light passing through the atmosphere to be 1.5 times the vertical thickness of the atmosphere. （AM1.5就是光线通过大气的实际距离为大气垂直厚度的1.5倍）

Blackbody Radiation (1900)

The Nobel Prize in Physics 1918
Max Planck

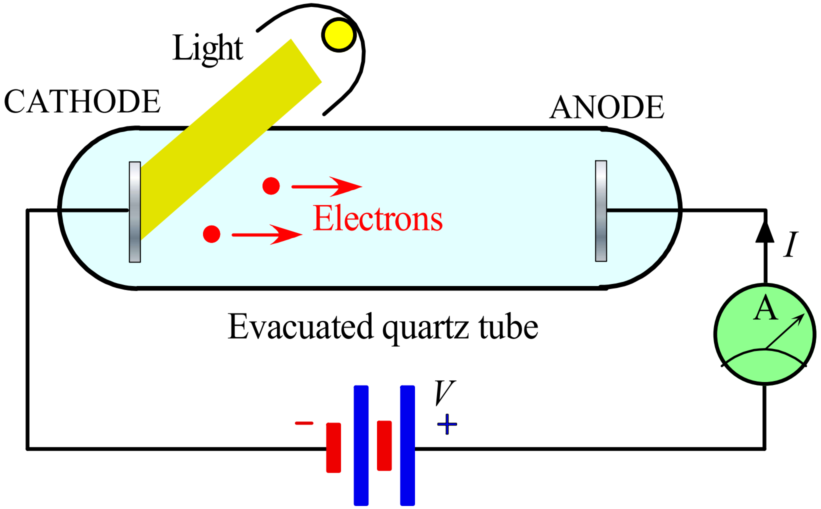
The Nobel Prize in Physics 1918

Max Karl Ernst Ludwig Planck

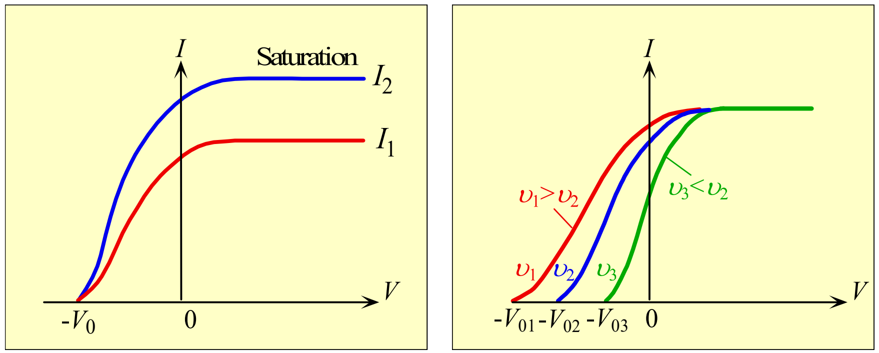
The Nobel Prize in Physics 1918 was awarded to Max Planck *"in recognition of the services he rendered to the advancement of Physics by his discovery of energy quanta"*.

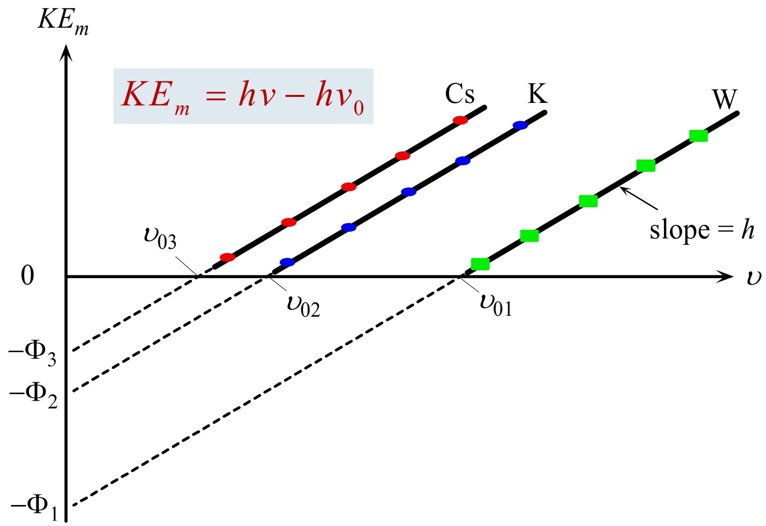
2. Photoelectric Effect (1905)

$$eV_0 = \frac{1}{2} m_e v^2 = KE_m$$



The Photoelectric Effect.





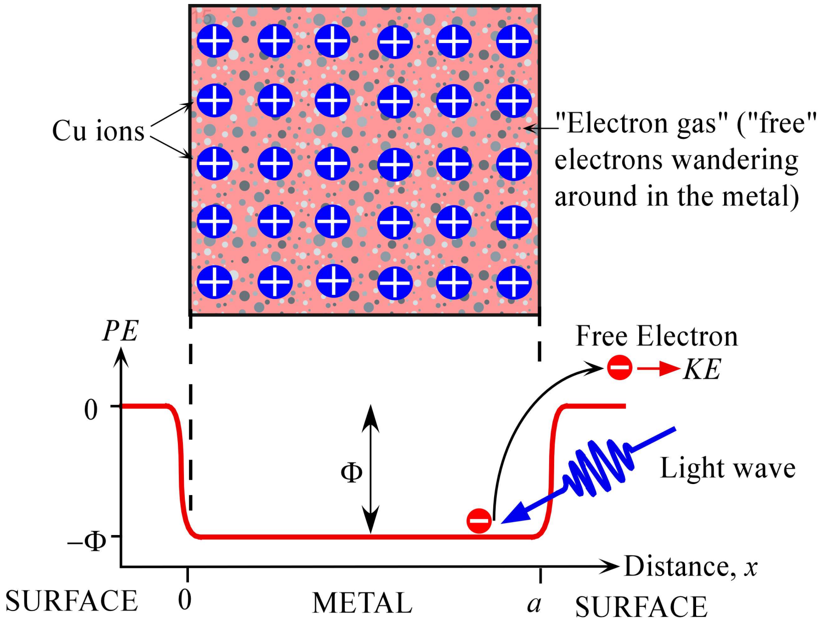
The effect of varying the frequency of light and the cathode material in the photoelectric experiment. The lines for the different materials have the same slope of h but different intercepts.

Electrons are ejected from the surface of a metal when exposed to light of certain frequency in a vacuum.

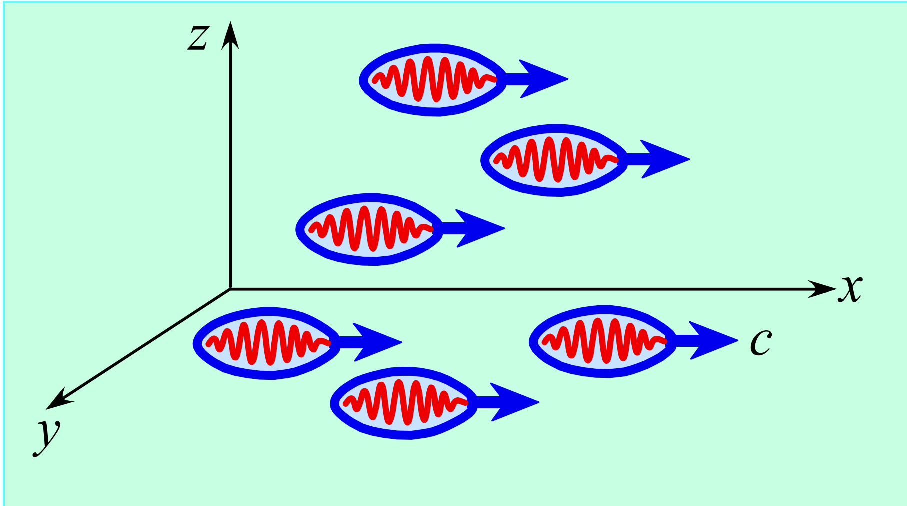
If the intensity of the light is large enough, the work function of the material will be overcome and an 'e' emitted from the surface independent of the incident frequency. However, the observed effect is that the maximum kinetic E of the photoelectron varies linearly with frequency with a limiting frequency $\nu = \nu_o$, below which no photoelectron is produced, as shown at the right.

http://www.kcvs.ca/site/projects/physics_files/photoelectric/photoelectricEffect.swf

Photoelectric Effect (1905)



The PE of an electron inside the metal is lower than outside by an energy called the workfunction of the metal. Work must be done to remove the electron from the metal.



Intuitive visualization of light consisting of a stream of photons (not to be taken too literally) [From © R. Serway, C. J. Moses and C.A. Moyer, *Modern Physics*, Saunders College Publishing, 1989, p.56, Fig. 2.16(b)]

Photon Flux: number of photons crossing a unit area per unit time.
光通量

$$I = \Gamma_{ph} h\nu$$

The photoemission condition

$$h\nu - \Phi \geq 0$$

work function: 功函数

Photoelectric Effect (1905)

EXAMPLE 3.1 **ENERGY OF A BLUE PHOTON** What is the energy of a blue photon that has a wavelength of 450 nm?

SOLUTION

The energy of the photon is given by

$$E_{\text{ph}} = h\nu = \frac{hc}{\lambda} = \frac{(6.6 \times 10^{-34}\ \text{J s})(3 \times 10^{8}\ \text{m s}^{-1})}{450 \times 10^{-9}\ \text{m}} = 4.4 \times 10^{-19}\ \text{J}$$

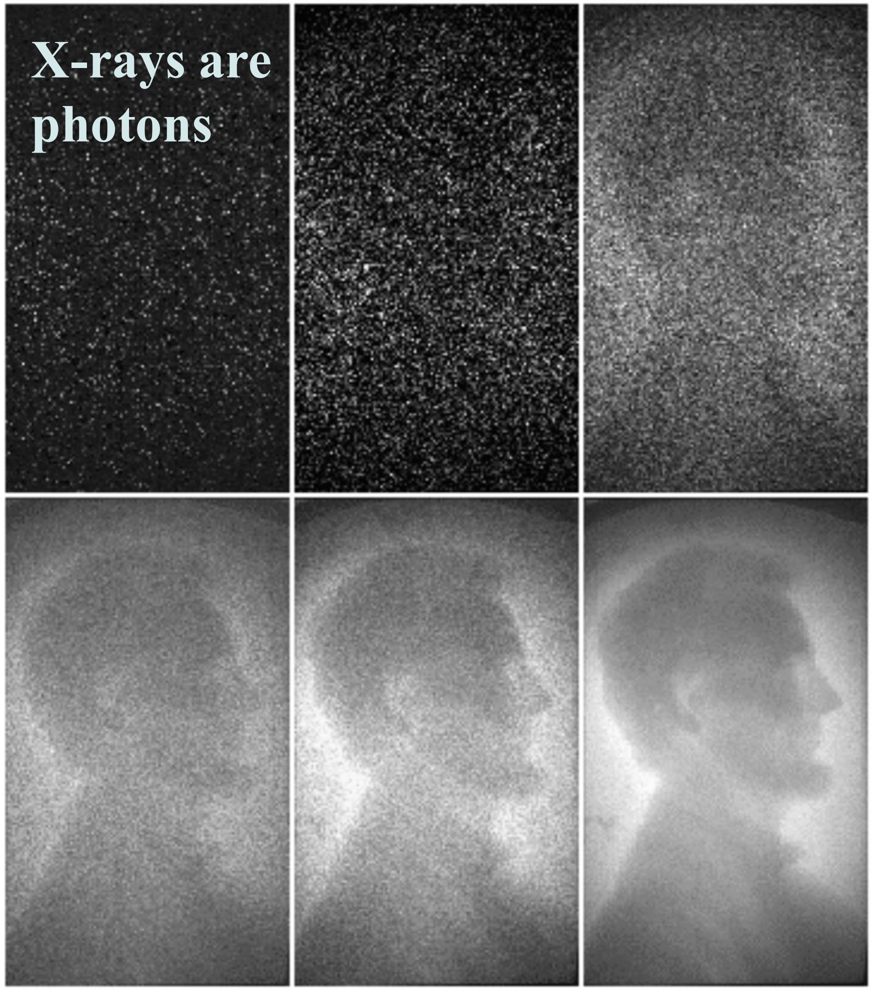
Generally, with such small energy values, we prefer electron–volts (eV), so the energy of the photon is

$$\frac{4.4 \times 10^{-19}\ \text{J}}{1.6 \times 10^{-19}\ \text{J/eV}} = 2.75\ \text{eV}$$

Photons

These electronic images were made with the number of photons indicated. The discrete nature of photons means that a large number of photons are needed to constitute an image with satisfactorily discernable details.

SOURCE: A. Rose, "Quantum and noise limitations of the visual process" *J. Opt. Soc. of America*, vol. 43, 715, 1953. (Courtesy of OSA.)



The Miracle Year 1905

The Miracle Year 1905 and Einstein's Publications (26 Yrs. old)
In One Year Einstein Publish Five Articles that Changed Our Understanding of the Science

① The Quantum Theory of Light

② Counting the Size of Atoms

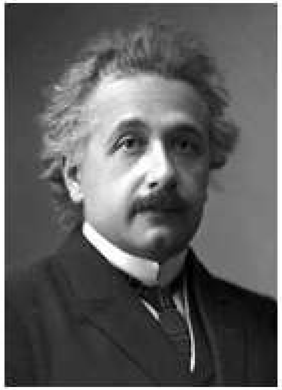
③ The Phenomenon of Brownian Motion

④ Special Relativity

⑤ Energy and Matter Equivalency

The Nobel Prize in Physics 1921
Albert Einstein

The Nobel Prize in Physics 1921



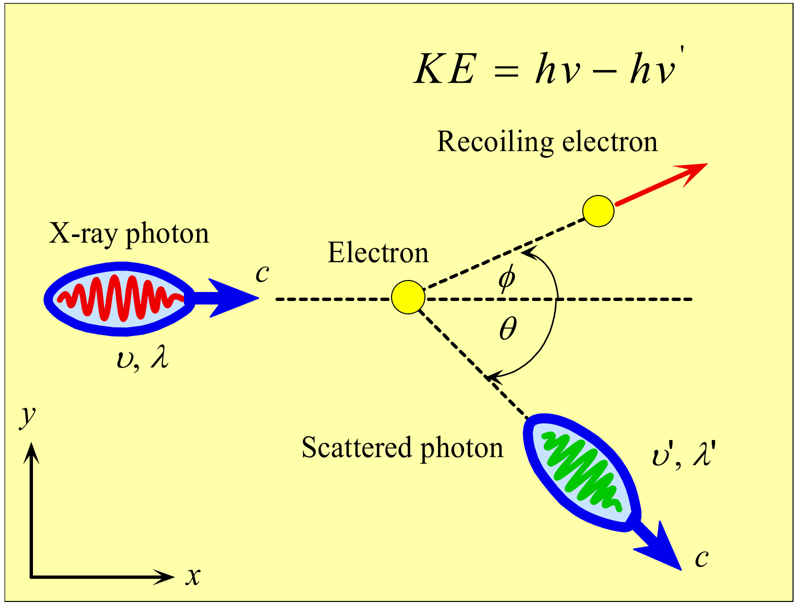
Albert Einstein

The Nobel Prize in Physics 1921 was awarded to Albert Einstein *"for his services to Theoretical Physics, and especially for his discovery of the law of the photoelectric effect".*

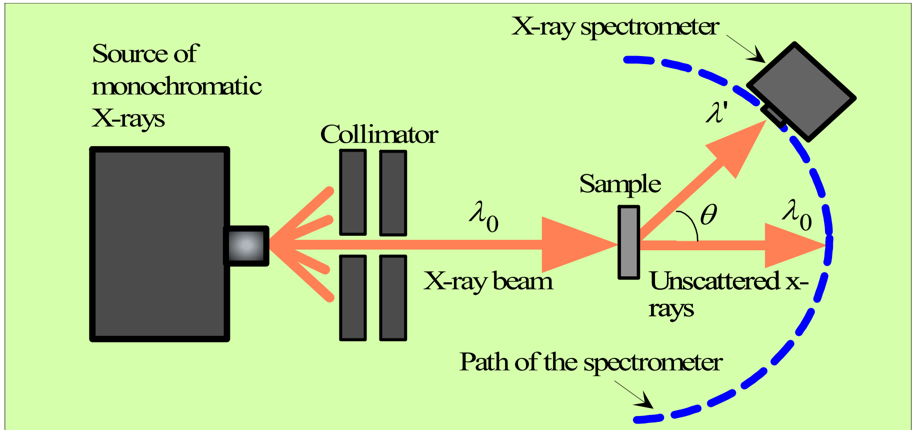
3. Compton Scattering

- Another direct evidence for the particle nature of radiation: Compton scattering
- Scattering of an X-ray photon by a "free" electron in a conductor: the radiation is scattered through a given angle with two components

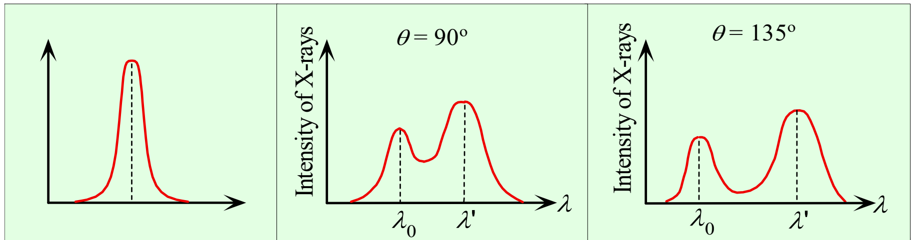
1. recoiling electron
2. scattered photon: wavelength shifted relative to the incident wavelength by an amount that depends on the angle



Scattering of an x-ray photon by a "free" electron in a conductor.



(a) A schematic diagram of the Compton experiment.



(b) Results from the Compton experiment

http://www.kcvs.ca/details.html?key=comptonScattering

Compton Scattering Vs Photoelectric Effect

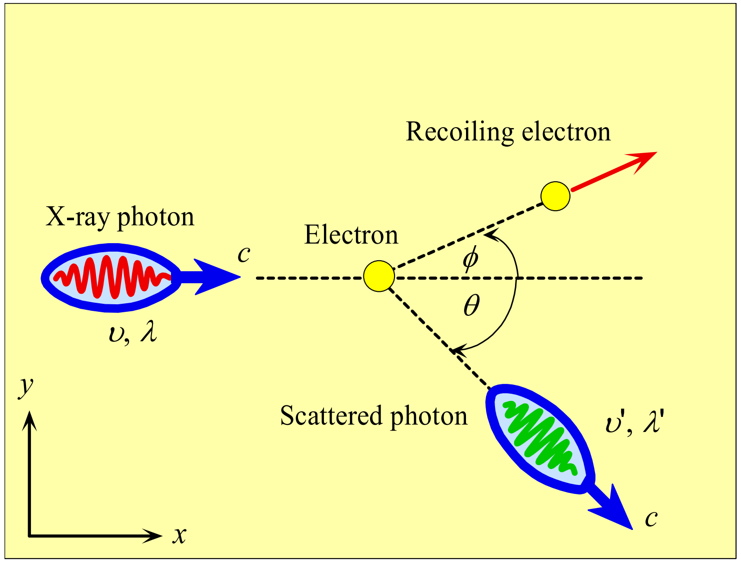
	Photoelectric Effect	Compton Scattering
Meaning	Electrons ejected out of atoms, photon is absorbed	Electrons are not ejected out of atoms. Incident photon is absorbed. A new photon of lower frequency is emitted
Energy	The entire energy of the incident photon is acquired by a single electron	The incident photon transfers only a part of its energy to an electron
Phenomenon	A low energy phenomenon, and the interacted photons disappear just after they deliver their energy to electrons	A mid-energy phenomenon, and the interacted photons are scattered by the electrons
Photon energy	The photon delivers its total amount of energy to a single electron	The photon transfers part of its energy to a single electron

Compton Scattering

Since the ‘e’ has a momentum p, we can accept that the X-ray also has momentum from the conservation of linear momentum. So the Compton exp. Showed that the momentum of the photon is related to its wavelength by

$$p = \frac{h}{\lambda}$$

$$E = h\nu$$



Scattering of an x-ray photon by a "free" electron in a conductor.

- Wave properties:
  $\nu, \lambda, Young's\ interference\ frings$

- Particle properties:
  p, photoelectric effect, Compton scattering

- Wave-particle duality
  Depend very much on the nature of the experiment

Compton Scattering

The Nobel Prize in Physics 1927
Arthur H. Compton, C.T.R. Wilson

The Nobel Prize in Physics 1927

Arthur Holly Compton

Charles Thomson Rees Wilson

The Nobel Prize in Physics 1927 was divided equally between Arthur Holly Compton *"for his discovery of the effect named after him"* and Charles Thomson Rees Wilson *"for his method of making the paths of electrically charged particles visible by condensation of vapour"*.

Summary

- Classical physics (Newton physics, Maxwell physics) cannot explain lots of experimental phenomena especially electron behaviors - when research go into submicron area.

  - ✓ Photoelectric effect, Campton scattering, Blackbody radiation
  - ✓ Light is a wave or a particle?
  - ✓ Electron is a particle or a wave?

- Using quantized photon concept, photoelectric effect, Campton scattering can be explained: light exhibit wave-particle duality, depending on the experiment and observation method, light can be treated as wave (interference and diffraction) or particle (momentum).

- Using quantized energy hv concept, Planck's law agree well with experimental observation.

- Experiment and explanation of Photoelectric effect, Campton scattering and Blackbody radiation open a new yet exciting era of quantum mechanics.